

Terraform-06

- 1). Watch the **Terraform-06** video.
- 2). Execute the script shown in the video.

Workspace:

```
* default
```

```
PS C:\Users\DELL\Desktop\Terraform> terraform workspace new project-A  
Created and switched to workspace "project-A"!
```

```
You're now on a new, empty workspace. Workspaces isolate their state,  
so if you run "terraform plan" Terraform will not see any existing state  
for this configuration.
```

```
PS C:\Users\DELL\Desktop\Terraform> terraform workspace list  
default  
* project-A
```

```
PS C:\Users\DELL\Desktop\Terraform> terraform workspace select default  
Switched to workspace "default".  
PS C:\Users\DELL\Desktop\Terraform> terraform workspace list  
* default  
project-A  
  
PS C:\Users\DELL\Desktop\Terraform>
```

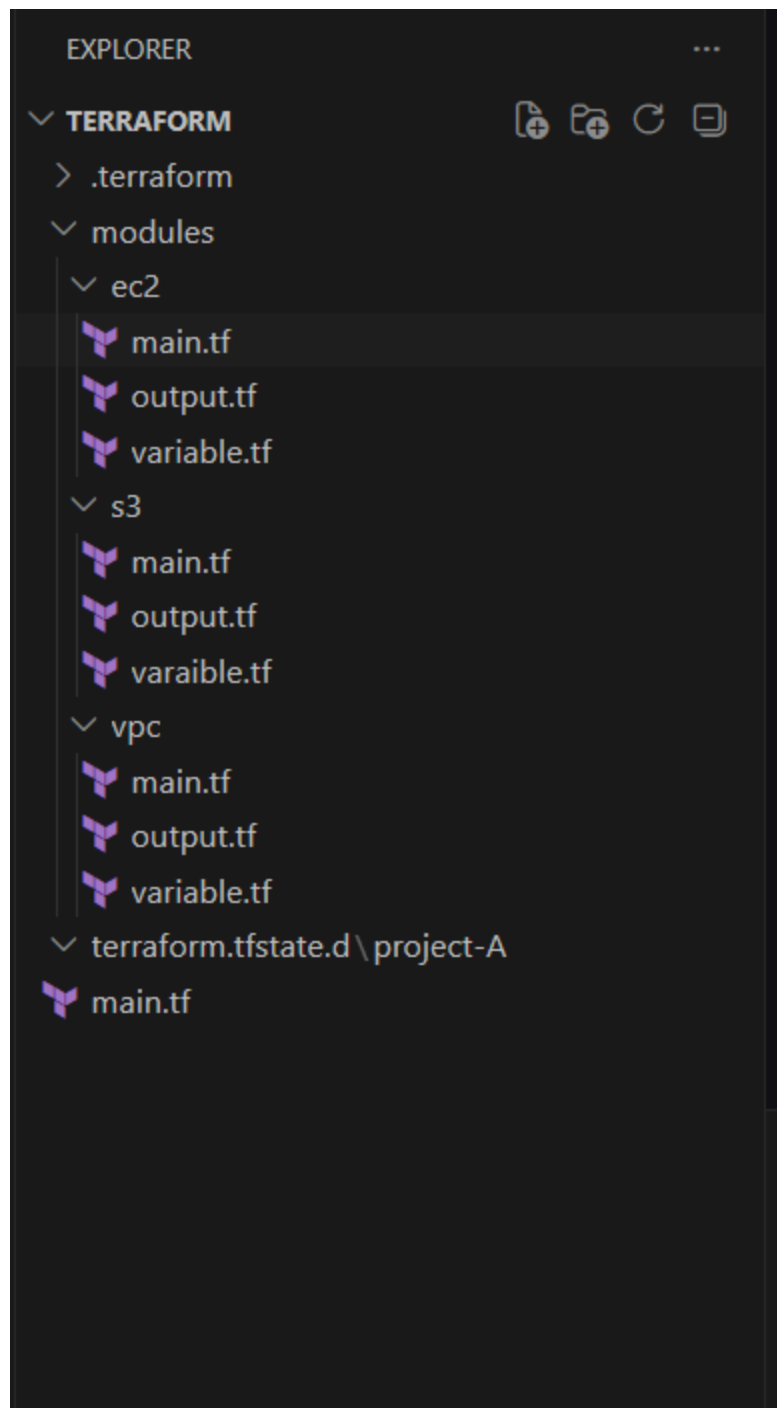
```
PS C:\Users\DELL\Desktop\Terraform> cd .\terraform.tfstate.d\  
PS C:\Users\DELL\Desktop\Terraform\terraform.tfstate.d> ls  
  
Directory: C:\Users\DELL\Desktop\Terraform\terraform.tfstate.d  
  
Mode                LastWriteTime         Length Name  
----                -  
d-----          6/17/2026 12:10 PM             project-A  
  
PS C:\Users\DELL\Desktop\Terraform\terraform.tfstate.d>
```

3). Provision **EC2, S3, and VPC** using **Terraform modules**.

I have created a terraform project in that created modules, in modules created

vpc, ec2 and s3

In terraform project main.tf and varibale.tf



In EC2 created main.tf

```
variable.tf \  output.tf \  main.tf ..\ec2 X  variable.tf ..\ec2  variable.tf X  ...  
modules > ec2 > main.tf > resource "aws_instance" "server"  
1 resource "aws_instance" "server" {  
2   ami           = var.ami_id  
3   instance_type = var.instance_type  
4   subnet_id     = var.subnet_id  
5  
6   tags = {  
7     Name = "Terraform-EC2"  
8   }  
9 }
```

```
modules > ec2 > variable.tf > variable "subnet_id"  
1 variable "ami_id" {}  
2 variable "instance_type" {}  
3 variable "subnet_id" {}
```

In s3 created main.tf

```
modules > s3 > main.tf > resource "aws_s3_bucket" "bucket"  
1 resource "aws_s3_bucket" "bucket" {  
2   bucket = var.bucket_name  
3 }
```

```
modules > s3 > variable.tf > variable "bucket_name"  
1 variable "bucket_name" {}
```

In vpc created main.tf

```
modules > vpc > main.tf > resource "aws_subnet" "public"  
1 resource "aws_vpc" "main" {  
2   cidr_block = var.vpc_cidr  
3  
4   tags = {  
5     Name = "terraform-vpc"  
6   }  
7 }  
8  
9 resource "aws_subnet" "public" {  
10  vpc_id     = aws_vpc.main.id  
11  cidr_block = var.subnet_cidr  
12  
13  tags = {  
14    Name = "public-subnet"  
15  }  
16 }
```

```
modules > vpc > variable.tf > variable "subnet_cidr"  
1 variable "vpc_cidr" {}  
2 variable "subnet_cidr" {}
```

Root main.tf:

```

main.tf > module "s3" > abc bucket_name
1  provider "aws" {
2    region = "ap-south-1"
3  }
4
5  module "vpc" {
6    source     = "./modules/vpc"
7    vpc_cidr   = "10.0.0.0/16"
8    subnet_cidr = "10.0.1.0/24"
9  }
10
11 module "s3" {
12   source     = "./modules/s3"
13   bucket_name = "bhargav-terraform-bucket-0291"
14 }
15
16 module "ec2" {
17   source       = "./modules/ec2"
18   ami_id       = "ami-0f58b397bc5c1f2e8"
19   instance_type = "t3.micro"
20   subnet_id     = module.vpc.subnet_id
21 }

```

Terraform init:

```

SUCCESS: The process with PID 3190 has been terminated.
PS C:\Users\DELL\Desktop\Terraform> terraform init
Initializing modules...
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.50.0...
- Installed hashicorp/aws v6.50.0 (signed by HashiCorp)

Initializing the backend...

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

```

Terraform plan:

Terraform apply:

only 'yes' will be accepted to approve.

```
Enter a value: yes
```

```
module.s3.aws_s3_bucket.bucket: Creating...
module.vpc.aws_vpc.main: Creating...
module.vpc.aws_vpc.main: Creation complete after 2s [id=vpc-02cf2974cc22ead3e]
module.vpc.aws_subnet.public: Creating...
module.s3.aws_s3_bucket.bucket: Creation complete after 2s [id=bhargav-terraform-bucket-0291]
module.vpc.aws_subnet.public: Creation complete after 0s [id=subnet-0eb5aebf084e6e753]
module.ec2.aws_instance.server: Creating...
module.ec2.aws_instance.server: Still creating... [00m10s elapsed]
module.ec2.aws_instance.server: Creation complete after 13s [id=i-0612c4c7d773ccd7c]
```

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.

```
PS C:\Users\DELL\Desktop\Terraform>
```

verify:ec2

Instances (1) Info

Last updated about 2 hours ago

Connect Instance state Actions Launch instances

Saved filter sets
Running Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	Terraform-EC2	i-0612c4cd7d773ccd7c	Running	t3.micro	Initializing	ap-south-1b	-

Select an instance

verify:vpc

Your VPCs

[VPCs](#) | [VPC encryption controls](#)

Your VPCs (2) [Info](#)

Last updated about 2 hours ago [Actions](#) [Create VPC](#)

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv...
☐	terraform-vpc	vpc-02cf2974cc22ead3e	Available	-	-	Off	10.0
☐	-	vpc-0a8ab810f0b1433db	Available	-	-	Off	172

Verify: s3

Buckets

[General purpose buckets](#) [All AWS Regions](#) [Directory buckets](#)

General purpose buckets (2) [Info](#)

[Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Find buckets by name

☐	Name	AWS Region	Creation date
☐	bhargav-terraform-bucket-0291	Asia Pacific (Mumbai) ap-south-1	June 17, 2026, 13:31:37 (UTC+05:30)
☐	my-terraform-state-bucket-0117	Asia Pacific (Mumbai) ap-south-1	June 16, 2026, 15:35:35 (UTC+05:30)

Account snapshot [Info](#)

Updated daily

[View dashboard](#)

Storage Lens provides visibility into storage usage and activity trends.

External access summary [Info](#)

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

4). Provision **EC2** for **3** different environments (Dev, Staging, and Prod) using **Terraform** workspaces.

main.tf:

```
main.tf \ X variable.tf \ output.tf \ main.tf ...\ec2 3 variable.tf ...\ec2 output.tf ...\ec2 main.tf
main.tf > resource "aws_instance" "server" > abc ami
1 provider "aws" {
2   region = "ap-south-1"
3 }
4
5 resource "aws_instance" "server" {
6   ami = "ami-0e38835daf6b8a2b9"
7   instance_type = lookup(
8     {
9       dev = "t3.micro"
10      staging = "t3.micro"
11      prod = "t3.micro"
12    },
13    terraform.workspace,
14    "t3.micro"
15  )
16
17  tags = {
18    Name = "${terraform.workspace}-server"
19    Environment = terraform.workspace
20  }
21 }
```

outputs.tf

```
output.tf > output "workspace"
1 output "instance_id" {
2   value = aws_instance.server.id
3 }
4
5 output "workspace" {
6   value = terraform.workspace
7 }
```

Terraform init:


```
PS C:\Users\DELL\Desktop\Terraform> terraform init
Initializing provider plugins found in the configuration...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.50.0

Initializing the backend...
Initializing provider plugins found in the state...
- Reusing previous version of hashicorp/aws
- Using previously-installed hashicorp/aws v6.50.0

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.
```

Create Workspaces

Dev:

```
PS C:\Users\DELL\Desktop\Terraform> terraform workspace new dev
Created and switched to workspace "dev"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
```

Select workspace:

Terraform apply:

```
PS C:\Users\DELL\Desktop\Terraform> terraform workspace select dev
PS C:\Users\DELL\Desktop\Terraform> terraform apply -auto-approve

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.server will be created
+ resource "aws_instance" "server" {
  + ami                  = "ami-0e38835daf6b8a2b9"
  + arn                  = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone     = (known after apply)
  + disable_api_stop     = (known after apply)
  + instance_id          = (known after apply)
  + workspace            = "dev"
aws_instance.server: Creating...
aws_instance.server: Still creating... [00m10s elapsed]
aws_instance.server: Creation complete after 13s [id=i-07ceb7ad091bbd470]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

instance_id = "i-07ceb7ad091bbd470"
workspace = "dev"
PS C:\Users\DELL\Desktop\Terraform>
```

verify:

Successfully initiated stopping of i-0612c4c7d773ccd7c

Instances (1) Info

Last updated 3 minutes ago

Connect Instance state Actions Launch instances

Saved filter sets Running Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	
☐	dev-server	i-07ceb7ad091bbd470	Running	t3.micro	Initializing	ap-south-1a	ec2-13-233-37-103.ap-...	1

stage:

```
PS C:\Users\DELL\Desktop\Terraform> terraform workspace new stage
Created and switched to workspace "stage"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.

PS C:\Users\DELL\Desktop\Terraform> terraform workspace select stage
PS C:\Users\DELL\Desktop\Terraform> terraform workspace list
default
dev
project-A
* stage
```

Terraform apply:

```
PS C:\Users\DELL\Desktop\Terraform> terraform apply -auto-approve

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.server will be created
+ resource "aws_instance" "server" {
  + ami               = "ami-0e38835daf6b8a2b9"
  + arn               = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone = (known after apply)
  + disable_api_stop  = (known after apply)
  + disable_api_termination = (known after apply)
```

```

Changes to Outputs:
  + instance_id = (known after apply)
  + workspace   = "stage"
aws_instance.server: Creating...
aws_instance.server: Still creating... [00m10s elapsed]
aws_instance.server: Creation complete after 13s [id=i-052fea0a8d4897c71]

```

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

```

instance_id = "i-052fea0a8d4897c71"
workspace   = "stage"
PS C:\Users\DELL\Desktop\Terraform>

```

verify:

Successfully initiated stopping of i-0612c4c7d773ccd7c

Instances (2) Info Last updated less than a minute ago Connect Instance state Actions Launch instances

Saved filter sets: Running Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	F
☐	stage-server	i-052fea0a8d4897c71	Running	t3.micro	Initializing	ap-south-1a	ec2-43-205-242-30.ap-...	4
☐	dev-server	i-07ceb7ad091bbd470	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-13-233-37-103.ap-...	1

prod:

Created and switched to workspace "prod"!

You're now on a new, empty workspace. Workspaces isolate their state, so if you run "terraform plan" Terraform will not see any existing state for this configuration.

```
PS C:\Users\DELL\Desktop\Terraform> terraform workspace select prod
```

```
PS C:\Users\DELL\Desktop\Terraform> terraform workspace list
```

```

default
dev
* prod
project-A
stage

```

Terraform apply:

```
PS C:\Users\DELL\Desktop\Terraform> terraform apply -auto-approve
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# aws_instance.server will be created
+ resource "aws_instance" "server" {
  + ami              = "ami-0e38835daf6b8a2b9"
  + arn              = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone = (known after apply)
  + disable_api_stop  = (known after apply)
  + disable_api_termination = (known after apply)
```

Changes to Outputs:

```
+ instance_id = (known after apply)
+ workspace   = "prod"
```

aws_instance.server: Creating...

aws_instance.server: Still creating... [00m10s elapsed]

aws_instance.server: Creation complete after 13s [id=i-03e804e87d4518aa5]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

```
instance_id = "i-03e804e87d4518aa5"
```

```
workspace = "prod"
```

```
PS C:\Users\DELL\Desktop\Terraform>
```

verify:

Instances (3) Info								
Last updated less than a minute ago Refresh Connect Instance state Actions Launch instances								
Saved filter sets: Running Find Instance by attribute or tag (case-sensitive)								
Instance state = running Clear filters								
☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	F
☐	stage-server	i-052fea0a8d4897c71	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-43-205-242-30.ap-...	4
☐	prod-server	i-03e804e87d4518aa5	Running	t3.micro	Initializing	ap-south-1a	ec2-13-206-196-26.ap-...	1
☐	dev-server	i-07ceb7ad091bbd470	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-13-233-37-103.ap-...	1